OMAR AHMED

Software Engineer | Node.js & TypeScript

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Portfolio | GitHub | LinkedIn

PROFILE

Software engineer with **3 years of experience** building Node.js and TypeScript backends across education, booking, nutrition, e-commerce and SaaS. Experienced in modular architecture, PostgreSQL performance, messaging and real-time systems, with React and Angular experience.

TECHNICAL SKILLS

Backend: Node.js, NestJS, Express.js, TypeScript, JavaScript, REST, GraphQL

Architecture & messaging: Hexagonal architecture, CQRS, DDD, microservices, RabbitMQ, WebSockets

Data: PostgreSQL, TypeORM, MongoDB, Redis, MySQL, SQL Server, Kysely

Frontend & delivery: React, Angular, Next.js, Docker, AWS, Git, Jest, Testcontainers, k6

EXPERIENCE

Raise Right | Node.js Developer

Jun 2025 - Present | Full-time | Cairo, on-site

- Developed services for **EduMarket, partner service management, a partner B2B marketplace, Roli transportation, central Admin and Crew operations**. Roli connects drivers, supervisors and parents, with partner-side transportation management.
- Contributed to EduMarket's **58-module NestJS backend**. In load tests with **300 concurrent users**, raised throughput **6.8x** (4.85 to 32.8 req/s), cut p95 latency from **60s to 9.9s**, and reduced HTTP failures from **79.5% to approximately 0%**.
- Diagnosed bottlenecks using **pg_stat_statements**; used targeted indexes, DataLoader batching and tag-based Redis caching to reduce top query times from **6-15 seconds to sub-millisecond**, with a measured **99.9% cache-hit rate**.
- Restored enrollment and wallet flows from **0% to 100% success under load**; integrated Kashier and Paymob behind circuit breakers. Removed **41 circular module dependencies** and introduced dependency checks before commits and pushes.
- Supported reliability with **3,000+ tests across 428 Jest/Testcontainers suites** and **23 k6 scenarios**, including a 20-to-200 RPS spike and a 45-minute soak at 15 RPS. The **1,000,000-user dataset was seeded test data**. Used Sentry, Prometheus and Loki for observability.

Apes Smart Solutions | Node.js Developer

Feb 2025 - Dec 2025 | Part-time | Remote

- Delivered backend services across **3 product areas**: nutrition applications, Spark activity bookings, and Views property/hospitality workflows.
- Applied **hexagonal architecture** and RabbitMQ messaging; used **TypeORM with PostgreSQL** for relational data and **MongoDB for audit records**.
- Built services handling **2,000+ requests/second at 99.9% uptime**; used RabbitMQ to reduce synchronous blocking by **approximately 70%** and supported **1,000+ daily bookings**.
- Improved logging and tracing to reduce incident debugging time by **60%**.

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EXPERIENCE & SELECTED WORK

Sarri Technology | MERN Stack Developer

Oct 2023 - May 2025 | Full-time | Cairo, on-site

- Modernized a **10,000-MAU** application, reducing critical bugs by **40%** and API latency by **45%**; built multi-tenant e-commerce, booking and HR SaaS workflows.
- Built **40+ REST APIs** serving **5M+ daily requests at 99.99% uptime**; rolled out **v2 and v3** with zero-downtime deployments.
- Built analytics dashboards processing **100M+ data points/month**; worked with an **8-person frontend team** to accelerate feature delivery by **25%**.
- Delivered reusable payment and notification components, team/payroll workflows, WebSockets updates and Google Maps integrations.

CustEx | Backend Developer

Aug 2023 - Nov 2023 | Part-time | Remote

- Built real-time interaction and voting services handling **15,000-20,000 messages/day** with **under 150ms delivery**; reduced connection latency by **25%** with **99.7% connection success**.

SELECTED PROJECTS

EduMarket

NestJS, TypeScript, PostgreSQL, Redis, GraphQL, Angular

Bilingual education marketplace; modular backend, partner and administration workflows, reporting and measured performance improvements.

Spark

NestJS, Node.js, MongoDB, Socket.io, React, Next.js

Activity booking services covering availability, notifications and integrations.

Views

NestJS, Node.js, Express, MongoDB, React

Property and hospitality backend workflows for listings, reservations, search and notifications.

Jam3ty

NestJS, Node.js, TypeScript, MongoDB, Angular

LMS serving Arab Open University students: courses, enrollment, content, progress and assessments.

ADDITIONAL TECHNOLOGIES

Python, FastAPI, Mosquitto, WebRTC, MQTT, AMQP, gRPC, Protocol Buffers; AWS S3, EC2, RDS, ECS, Lambda, Amplify and SQS. Agile development practices.

EDUCATION

Fayoum University | B.Sc. Computer Science, Artificial Intelligence

Aug 2019 - Jun 2023